

BONGIMUSA SANDILE KHOZA

Software Developer | AI & Machine Learning Engineer | Data Analyst | Cybersecurity & IT Risk | QA & Testing

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South African Citizen | Driver's Licence: Code 10 (C1) | Immediately Available | Willing to Relocate

PROFESSIONAL SUMMARY

Results-driven BSc Computer Science and Mathematics graduate (NQF Level 7) from North-West University with practical experience developing and deploying software applications across software engineering, artificial intelligence, machine learning, backend development, data analytics, cybersecurity, and cloud technologies. Proficient in Python, JavaScript, Java, SQL, Node.js, HTML, CSS, REST APIs, Git, and modern AI frameworks, with experience building end-to-end solutions involving computer vision, natural language processing, data visualization, authentication, and database design. Recognized for strong analytical thinking, problem-solving, attention to detail, and the ability to quickly learn and apply new technologies. Passionate about building innovative, scalable, and user-focused solutions while contributing effectively within collaborative Agile teams. Immediately available and willing to relocate.

TECHNICAL SKILLS

Languages & Development: Python, JavaScript, Java, C++, SQL, HTML5, CSS3, Node.js, Streamlit — OOP, Data Structures & Algorithms, REST APIs, SDLC, Agile/Scrum

Backend & APIs: Node.js, Express.js, MongoDB, Mongoose, JWT, bcrypt, Swagger, PostgreSQL

AI, ML & Computer Vision: Scikit-learn, DeepFace, KMeans Clustering, Neural Networks, Deep Learning, NLP, Computer Vision, Face Recognition, TextBlob

LLMs & Generative AI: LLaMA 3.3 70B (Groq API), Gemini 2.5 Flash, Claude AI, Prompt Engineering, Azure AI Engineer, Azure Cognitive Services

Data Analytics & BI: Pandas, NumPy, Matplotlib, Plotly, Power BI, MS Excel (VLOOKUP, Pivot Tables, Macros, Dynamic Dashboards), SQL, Statistical Analysis, Financial Modelling

QA, Testing & IT Risk: Functional & System Testing, Test Plan & Test Case Design, Defect Tracking, Root Cause Analysis, IT Risk Assessment, Control Gap Identification, Audit Documentation, BABOK Awareness

Cybersecurity & Compliance: Cybersecurity Fundamentals, POPIA, RBAC, Network Security, DHCP Snooping, Security Testing, Incident Management, ISO 9001 / COBIT / NIST Awareness

Networking & Infrastructure: Cisco Packet Tracer, VLANs, DHCP (IPv4/IPv6), Static NAT, TCP/IP, Protocol Analysis, Inter-VLAN Routing, Linux (Basic)

DevOps & Tools: Git/GitHub, VS Code, Jupyter Notebook, Google Colab, Streamlit Cloud, Render, GitHub Pages, MS Office 365

LIVE DEPLOYED PROJECTS

SafeRide — AI Face Recognition Ride Safety Platform | *Python · Streamlit · DeepFace · Gemini 2.5 Flash · Streamlit Cloud* | 2026
saferide-rrwv43c8efmxhtbmu2xgjr.streamlit.app | github.com/SANDILE19991111/saferide

- Engineered an AI safety platform using DeepFace computer vision for live biometric passenger identity verification, directly applicable to cybersecurity, IAM, and fintech security domains.
- Built face recognition sign-up/sign-in with confidence scoring, dynamic ride pricing (R12-R25/km with peak surcharges), SOS emergency location logging, AI-generated safety briefings via Gemini 2.5 Flash, and a full ride-history dashboard.
- Resolved real-world deployment challenges including DeepFace installation on cloud infrastructure, secure API key management via Streamlit Secrets, and camera/session state handling.

SAPS iLodge — AI-Powered Complaint & Ticket Management Platform | *Node.js · Express · Groq API (LLaMA 3.3 70B) · JavaScript · Render* | 2026
saps-ilodge-week7.onrender.com | github.com/SANDILE19991111/saps-ilodge

- Built a full-stack complaint management platform with role-based access for Citizens, Staff, and Administrators.

- Integrated an LLM-powered keyword routing engine to auto-assign tickets to departments with a confidence score, plus an AI chatbot for citizen support.
- Developed analytics dashboards, predictive workload forecasting, automated executive reporting, and a workflow automation layer covering SLA breach alerts and approval queues.

DeskFlow — Internal IT Ticketing REST API | *Node.js · Express · MongoDB · JWT · Swagger* | 2026

github.com/SANDILE19991111/deskflow

- Developed a secure REST API for internal IT ticket management with server-enforced role-based access control.
- Implemented JWT authentication, bcrypt password hashing, request validation (express-validator), and centralised error handling normalising Mongoose/JSON errors into proper HTTP responses.
- Documented all endpoints with Swagger and verified functionality via a Postman test collection.

ContentBot (Email Drafter) — LLM-Powered Email Automation | *Node.js · Express · Groq API · LLaMA 3.3 70B · Render* | 2026

email-drafter-ciay.onrender.com | github.com/SANDILE19991111/email-drafter

- Built and shipped an AI email generation tool integrating LLaMA 3.3 70B via Groq API, supporting multiple tones (Professional, Friendly, Casual, Persuasive) with AI-generated subject lines.
- Implemented system-prompt engineering to govern AI behaviour, secure API key handling, and an automated deployment pipeline on Render.

SentimentIQ — AI Sentiment Analysis Dashboard | *Python · Streamlit · TextBlob · Plotly* | 2026

sentiment-tool-hd6pd9vitvfbfn89jjakf.streamlit.app | github.com/SANDILE19991111/sentiment-tool

- Developed an NLP dashboard classifying customer reviews and social comments as Positive, Negative, or Neutral using polarity scoring.
- Built interactive bar/pie/donut charts, positive/negative word clouds, a polarity-distribution histogram, and CSV/Markdown export of results.

OceanIQ — AI Blue Economy Assistant Chatbot (Team Project) | *HTML · CSS · JavaScript · Prompt Engineering · GitHub Pages* | 2025

sandile19991111.github.io/OceanIQ-Chatbot | github.com/SANDILE19991111/OceanIQ-Chatbot

- Served as Team Lead: directed frontend development, testing, and deployment for a 4-person team.
- Built an AI chatbot delivering role-based guidance (fishers, consumers, environmentalists, government officials) on sustainable marine practices and ocean conservation.
- Implemented accessibility features including voice input, adjustable text size, and high-contrast mode.

Enterprise Network Infrastructure Design | *Cisco Packet Tracer · VLANs · DHCP · Static NAT · TCP/IP* | 2025

github.com/SANDILE19991111/Academic-Project-Network-Configuration-Design

- Designed five independent network topologies (bus, ring, star, mesh, hybrid) and a hybrid multi-faculty network combining IPv4/IPv6 addressing, DHCP/DHCPv6, DHCP snooping, and VLAN segmentation with Router-on-a-Stick and port security.
- Configured Static NAT for controlled public access to internal web/DNS resources; diagnosed and resolved connectivity and security issues, restoring full connectivity across all topologies.

ACADEMIC & BOOTCAMP PROJECTS

Clinic Appointment System — Full-Stack Agile Delivery & QA | *JavaScript · SQL · PostgreSQL · Agile* | Apr – May 2025

- Designed a 4-entity normalised relational database (ERDs, SQL DDL/DML) and built a JavaScript frontend with RBAC and POPIA-compliant security controls.
- Executed 15+ functional test cases, identified 5+ control gaps, and produced audit-style working papers demonstrating QA and IT risk competency.
- Delivered on time within an Agile team of four; facilitated a stakeholder review presenting findings, risks, and corrective recommendations.

KMeans Clustering — Machine Learning Data Pipeline | *Python · Scikit-learn · NumPy · Matplotlib* | Mar – May 2025

- Analysed a 300+ user dataset, achieving 98% clustering accuracy (ARI: 0.3440); produced four visualisations and a structured analytical report with actionable insights.

Image Classification — Neural Network Model | *Neural Networks · Google Teachable Machine* | Feb 2025

- Trained a neural network on 100+ images across two classes; evaluated generalisation on unseen data and proposed a retail automation application for the model.

Database, Budgeting & Business Planning | *MS Excel · SQL · ERD | Apr – Jun 2022*

- Automated a payroll database for 15+ staff with zero discrepancies; built a 12-month dynamic financial budget dashboard and a procurement proposal with cost analysis, asset listing, and Gantt scheduling.

EDUCATION

Bachelor of Science: Computer Science and Mathematics — NQF Level 7 | *North-West University, Mafikeng | Jan 2022 – Dec 2025*

- Relevant coursework: Artificial Intelligence, Machine Learning, Databases, Software Engineering, Data Structures & Algorithms, OOP, Computer Networks, Operating Systems, Cybersecurity Fundamentals, Statistics, Linear Algebra, Calculus.
- Applied Python, Java, C++, JavaScript, and SQL across AI, systems, data, and network projects; completed full SDLC project lifecycles and Agile team delivery with documentation, sprint planning, and stakeholder reporting.

National Certificate: N3 — Engineering Studies | *Ekurhuleni West College | 2019*

National Senior Certificate — Bachelor's Pass | *Mchaka High School | Dec 2018*

CERTIFICATIONS & PROFESSIONAL TRAINING

- Full Stack Development Certificate — FNB App Academy (Jul 2025)
- Generative AI: Prompt Engineering Basics — IBM / Coursera (May 2026)
- Introduction to Artificial Intelligence — IBM / Coursera (May 2026)
- AI For Everyone — DeepLearning.AI / Coursera (Apr 2026)
- Introduction to Generative AI — Google Cloud / Coursera (May 2026)
- Python for Data Science, AI & Development — IBM / Coursera (2026)
- AI Essentials — Coursera (Jun 2026)

ADDITIONAL INFORMATION

Citizenship: South African

Driver's Licence: Code 10 (C1) — valid

Languages: English (Fluent), Xitsonga (Home Language), IsiZulu (Conversational)

Availability: Immediately available, willing to relocate

REFERENCES

Available upon request.